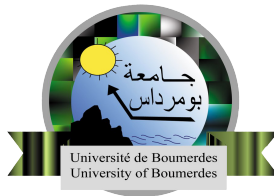


# Bachelor Project Title



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# Abbreviations

|       |                                   |
|-------|-----------------------------------|
| AUV   | Autonomous Underwater Vehicle     |
| DDR   | Differential Drive Robot          |
| DOF   | Degrees Of freedom                |
| FWMMR | Four Wheeled Mecanum Mobile robot |
|       | Global Positioning System         |
| DEV   | DIRECT ENOVATION VECTOR           |

# Chapter 1

## Introduction

# Chapter 2

## Theoretical Background

# Chapter 3

## Hardware system design

### 3.1 Introduction to 5G NR

Developed by the Third-Generation Partnership Project (3GPP), 5G New Radio (NR) represents the standardized air interface for fifth-generation mobile networks. This new radio access technology introduces significant enhancements over its predecessor, LTE, including:

Operation in higher frequency spectrum bands, enabling substantially wider transmission bandwidths and consequently supporting greater data throughput

An optimized ultra-lean architecture that minimizes persistent signaling, thereby improving energy efficiency while reducing network interference

Built-in forward compatibility that ensures adaptability to future technological developments through flexible allocation of time and frequency resources

Dramatically reduced latency, achieving transmission delays as low as 1 millisecond

Comprehensive implementation of beamforming and massive MIMO technologies across both control and user plane communications .

[1].

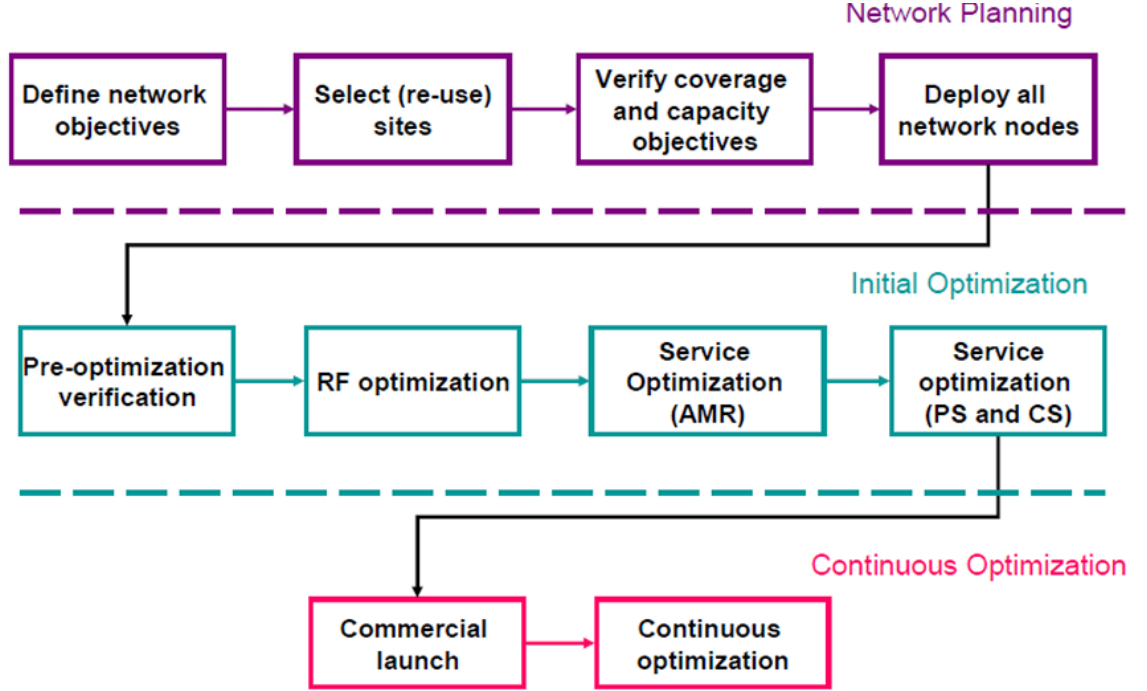


Figure 3.1 – General network deployment steps

## 3.2 Network planning and optimization

Network planning and optimization is an iterative process involving topological design, network realization, and network synthesis. Its primary goal is to ensure that both newly deployed and existing cellular networks meet user demands and operator (MNO) requirements.

Historically, planning and optimization were treated as separate processes (see Figure 4.1). However, advancements in Radio Access Technologies (RAT) and the introduction of New Radio (NR) concepts in 5G networks have blurred the traditional boundaries between them. Studies [80][81] argue that planning and optimization are deeply interconnected and must be addressed jointly, particularly in ultra-dense and heterogeneous networks (HetNets). Effective planning ensures balanced traffic distribution across base stations, which is critical for network performance. [2].

[3]

### 3.3 5G NR Use Cases

Unlike LTE, 5G New Radio was designed from inception to support three primary application scenarios:

Enhanced Mobile Broadband (eMBB): Representing the natural progression of mobile networks, eMBB delivers significantly higher throughput and improved user experiences. This use case supports applications requiring substantial data rates and capacity, primarily serving human-centric communication needs.

Massive Machine-Type Communications (mMTC): Catering to machine-centric applications, mMTC connects vast numbers of low-power devices such as sensors, meters, and monitoring equipment. These connections typically transmit minimal data and prioritize energy efficiency over high bandwidth.

Ultra-Reliable and Low Latency Communications (URLLC): Focused on mission-critical applications, URLLC enables machine-to-machine communication with exceptional reliability and sub-millisecond latency. Key applications include vehicle-to-everything (V2X) systems, autonomous vehicles, industrial automation, and remote healthcare services . [4].

### 3.4 Atoll for 5G Network Planning and Optimization

Forsk's Atoll is a comprehensive wireless network planning platform that supports current and emerging technologies, including 5G New Radio (NR), LTE, NB-IoT, UMTS, and GSM. The software incorporates advanced features such as MIMO antenna systems, 3D beamforming, and millimeter-wave propagation modeling, enabling network operators and equipment vendors to effectively design, optimize, and plan next-generation wireless networks.

As mobile technology evolves, user expectations for network performance and service quality continue to rise. Atoll provides the necessary tools to meet these increasing demands through its advanced simulation and planning capabilities. The technical

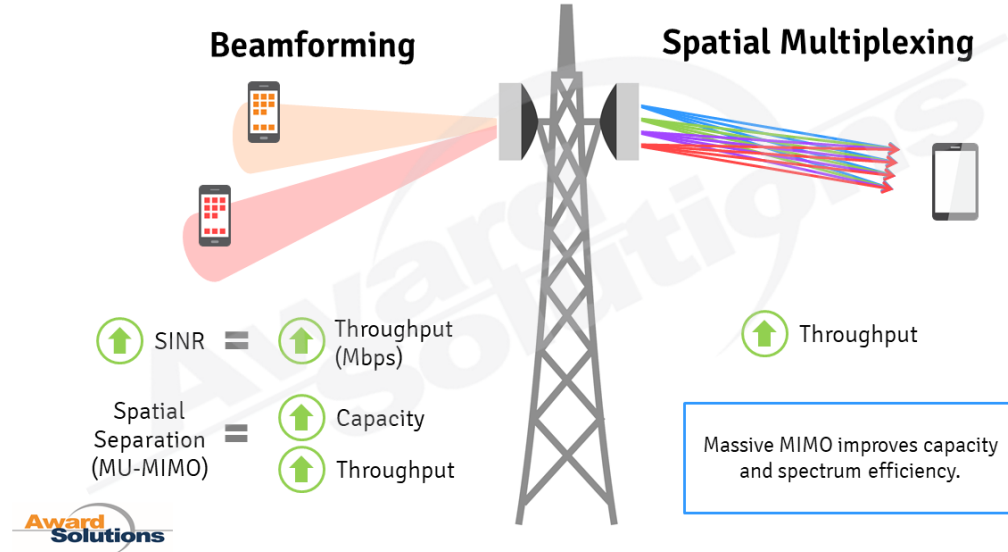


Figure 3.2 – *beamforming-spatial-multiplexing-award-solutions*

information presented in this section is primarily sourced from the official Atoll user manual and the Atoll Technical Reference Guide. [5].

### 3.5 Network Configuration in Atoll

Atoll's network design module enables comprehensive modeling of both equipment and operational parameters. The configuration architecture comprises three fundamental layers:

Site Configuration: Defines base station locations with adjustable parameters including:

Site identification name

Geographic coordinates (x/y position)

Elevation data (altitude)

Transmitter Configuration: Offers extensive parameterization through:

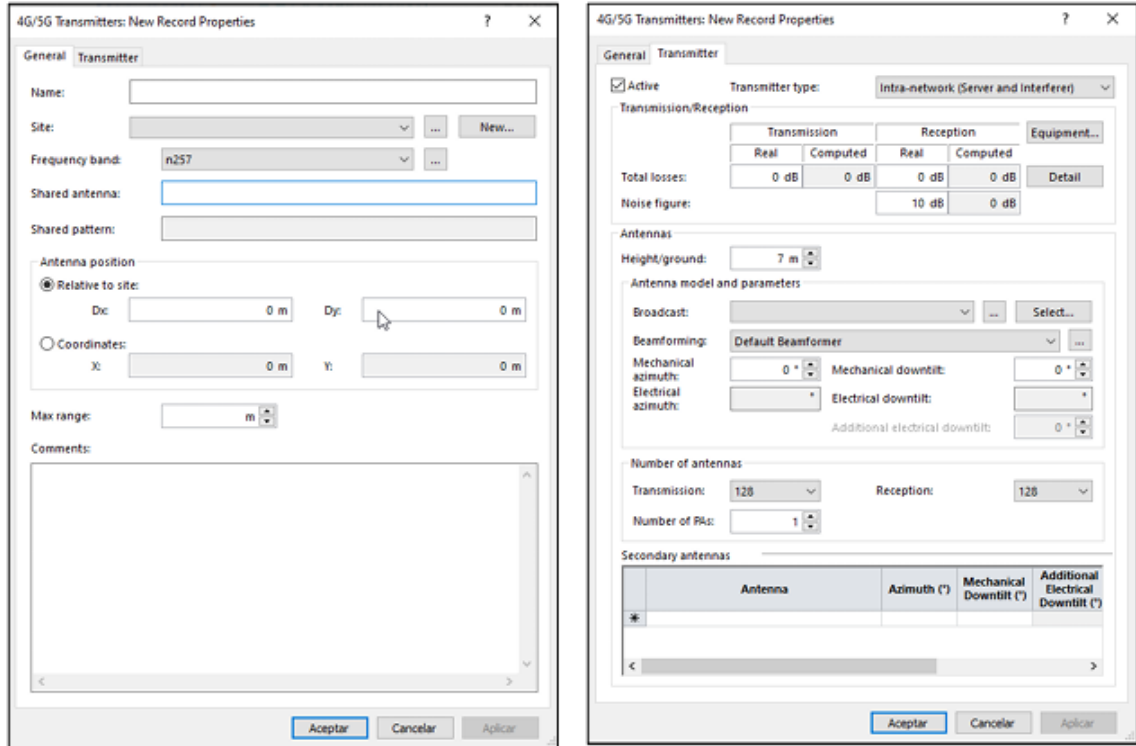


Figure 3.3 – Figure 2. New transmitter's general tab.

Figure 3.4 – Figure 3. New transmitter's transmitter tab

General settings (nomenclature, parent site association, frequency band selection, antenna positioning)

Transmission/reception loss specifications for:

Tower-Mounted Amplifier (TMA) systems

Feeder cable installations

Transceiver hardware components

Cell Configuration: Manages radio frequency coverage characteristics . [6].

### 3.5.1 propagation probabilities keys

In the antennas section, either a broadcast antenna or a beamforming antenna can be selected. If both are defined, only the broadcast antenna is used for path loss

calculation. The antenna height, number of antennas, mechanical azimuth, and downtilt can also be configured.

For cell configuration, multiple options are available. Each transmitter can support one or more cells of different technologies. Within each cell, key configuration parameters include:

Name: Default naming follows the transmitter name with an added suffix in parentheses.

Carrier: The cell's carrier frequency within the band.

Physical Cell ID (PCI): An integer value between 0 and 1007.

Max Power (dBm): The cell's maximum transmission power.

Layer: The network layer determining the serving cell.

Cell Type: Defines the cell as primary (PCell) or secondary (SCell for UL/DL).

Cell Individual Offset (CIO) (dB): Adjusts cell ranking for selection in 5G NR connected mode.

Cell Selection Threshold (dB): Modifies the Min SS-RSRP threshold for priority-based cell selection in 5G NR.

SS/PBCH Numerology: Numerology applied to SS/PBCH signals.

Traffic Numerology: Numerology used for traffic channels (PDCCH, PDSCH, PUSCH).

Diversity Support (DL): Downlink antenna diversity (none, transmit diversity, SU-MIMO, MU-MIMO).

Diversity Support (UL): Uplink antenna diversity (none, receive diversity, SU-MIMO, MU-MIMO).

Number of MU-MIMO Users (DL/UL): Average users sharing downlink/uplink resources in MU-MIMO.

Number of Users (DL/UL): Connected users in downlink/uplink.

| Propagation Class                                     | Description                                                                  | Calculation                                                                                                                                                                                                           |
|-------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Open                                                  | All open areas, streets, squares, flat parks, etc.                           | Open space, radio signals propagate freely.                                                                                                                                                                           |
| Water                                                 | Sea, rivers, lakes, etc.                                                     |                                                                                                                                                                                                                       |
| Elevated Open                                         | Elevated areas.                                                              | Filled obstruction area for radio signals.<br>Reception done on top.                                                                                                                                                  |
| Trees                                                 | Trees, forests, etc.                                                         | Deterministic vegetation.<br>Radio signals can go through with some loss, and are also diffracted and diffused.                                                                                                       |
| Building type 1<br>Building type 2<br>Building type 3 | Three types of buildings are defined to model different propagation effects. | Deterministic buildings that create an obstruction on which radio signals reflect and diffract. Radio signal levels are not calculated in a direct way but interpolated from signal levels in surrounding open areas. |
| Bridge                                                | Bridges, highways, etc.                                                      | Radio propagation is calculated on top of bridges. Radio signals can propagate below bridges.                                                                                                                         |

Figure 3.5 – Table 4. Deterministic propagation classes.

Traffic Load (DL/UL) (

Max Traffic Load (DL/UL) (

PRACH RSIs: Logical PRACH root sequences assigned to the cell.

Neighbours: Configurable intra- and inter-technology neighbors via a dedicated dialog box. [7].

section3GPP Model propagation network

| Propagation Class | Description                                                       | Calculation                                                                                                                                                           |
|-------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mixed Vegetation  | Parks or areas with a mixture of trees and open areas             | Radio signals can go through with some loss and diffract vertically. The clearance area is considered at the receiver location and reception is done at ground level. |
| Sparse Buildup    | Sparse building areas with a mixture of buildings and open areas. |                                                                                                                                                                       |
| Dense Buildup     | Dense building areas with a mixture of buildings and open areas.  |                                                                                                                                                                       |

Figure 3.6 – Table 5. Statistical propagation classes.

### 3.6 Predictions and simulations

Atoll allows to make different types of coverage predictions in order to analyse the signal levels, the signal quality and the network capacity. Atoll determines the serving cell for each pixel using the standard cell selection mechanism. Then, depending on the prediction definition, it calculates the required effective signal or parameter. Pixels are coloured if the display threshold condition is fulfilled.

[8].

# Chapter 4

## Software system design

# Chapter 5

## Simulation results

# Chapter 6

## Conclusion

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